

Football Match Manager - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager is a web-based tracking platform enabling users to browse, filter, and follow football matches in real-time. The system acts as a data aggregator, consuming third-party sports APIs to persist match states and user-specific interest mappings. Its core value lies in delivering high-freshness score updates and a personalized 'followed matches' experience for authenticated users.

1.2 Maturity Assessment

The specification is logically sound regarding user journeys but lacks a strategic foundation. The absence of a 'Goals & Objectives' section represents a critical structural gap that hinders the alignment of success KPIs with business value. While the functional requirements are granular, the project requires REFINEMENT to resolve uncertainties regarding the specific API provider and the final authentication protocol before execution.

1.3 Technical Stack

- **External Data Source**: Sports Data API (Mandatory)
- **Authentication**: Third-party OAuth Providers (Google, Facebook, Apple)
- **Frontend**: Responsive Web Design (Mobile & Desktop)

1.4 Architectural Constraints

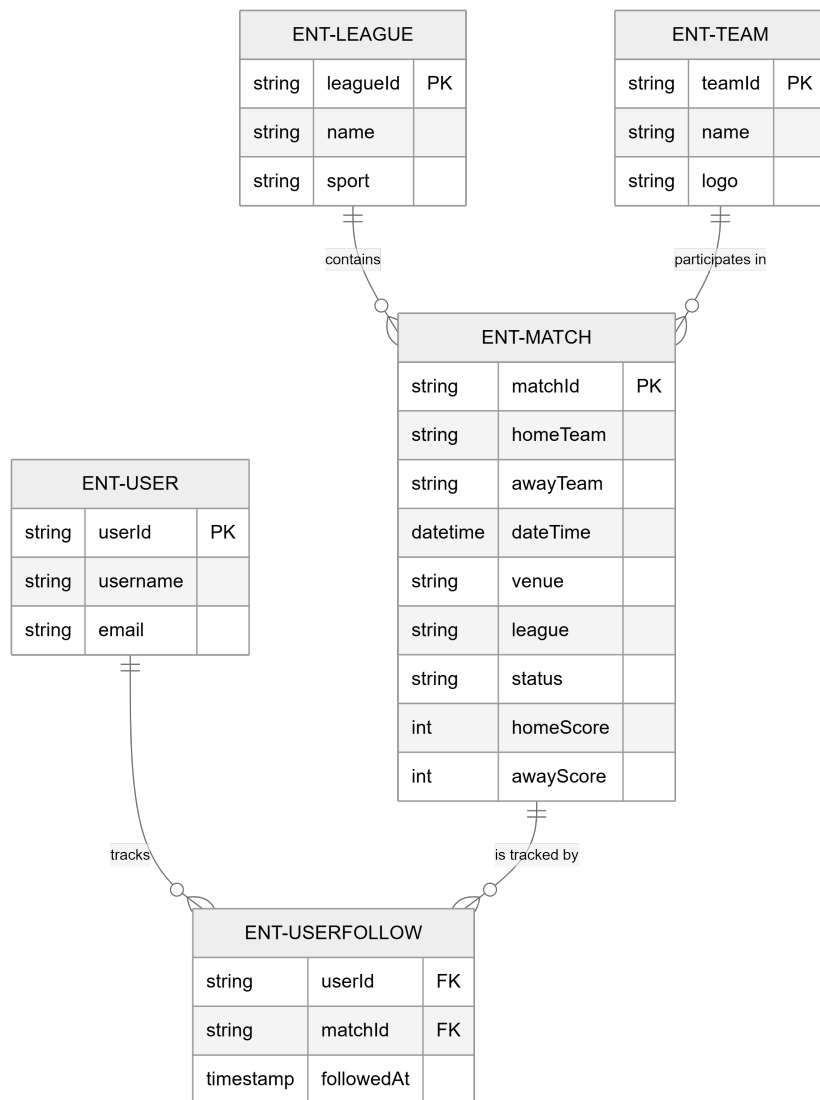
- **Data Refresh Intervals**: Upcoming match data must be refreshed at least every 6 hours.
- **Live Update Frequency**: Live score updates must occur at least every 1 minute during active matches.
- **Completion Latency**: Final scores must be updated within 5 minutes of match completion.
- **Performance Thresholds**: Match list load time < 2 seconds (95% requests, cached); Detail page load time < 1.5 seconds (95% requests).
- **Data Accuracy**: Live score delay must be < 60 seconds.
- **Scope Exclusions**: Push notifications, social features (sharing, commenting), and predictive analytics are strictly out-of-scope for MVP.

1.5 Critical Dependencies

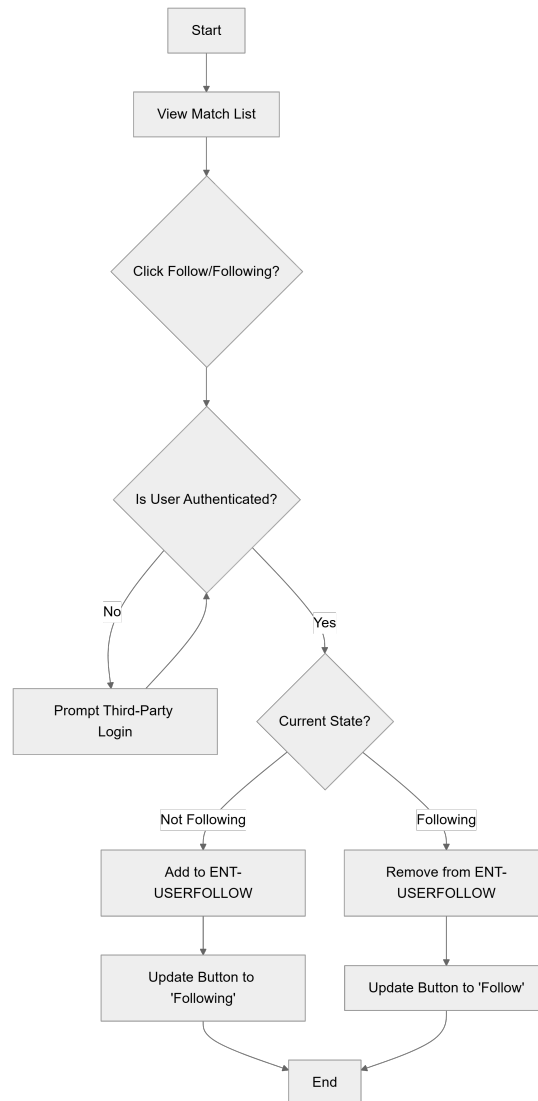
- **External Football Data API**: Mandatory for match schedules, live scores, and team/league metadata.
- **Third-party Authentication Provider**: Required for user session persistence and "Follow" functionality.
- **UserFollow Entity**: Strict relational join dependency between User and Match entities.
- **Referential Integrity**: UserFollow records must depend on existing User and Match IDs.

2. Architecture Workflows & Visual Diagrams

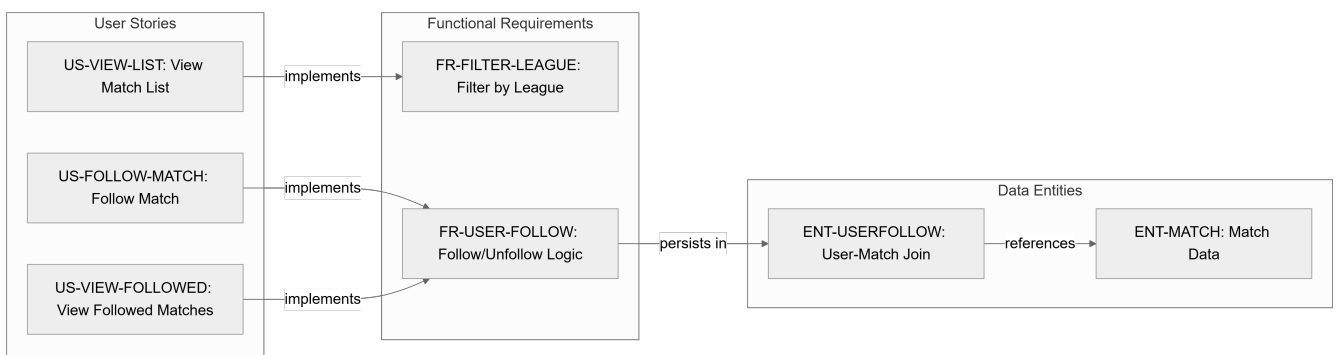
2.1 Data Model (ER Diagram)



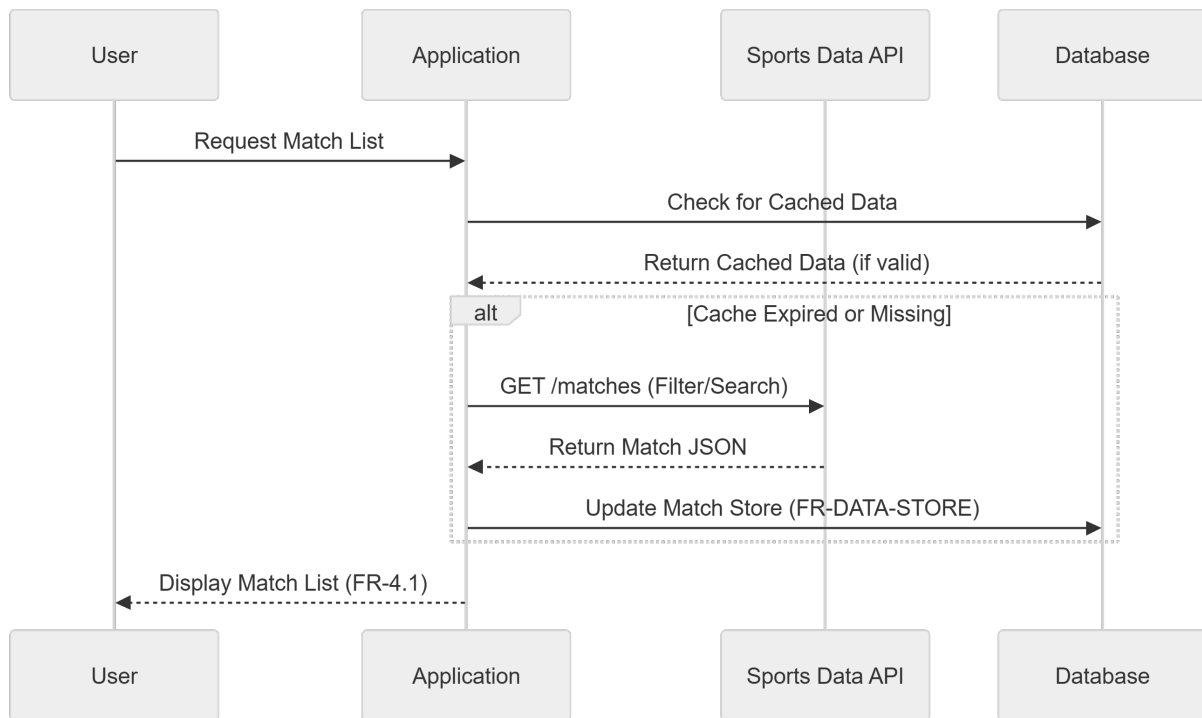
2.2 Match Following Workflow



2.3 Requirements Traceability Matrix



2.4 Match Data Retrieval Sequence



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Description	Source Section
US-VIEW-LIST	User Story	View upcoming football matches to decide which to follow	Scenario 1: Viewing Match List
US-FOLLOW-MATCH	User Story	Mark a match as interesting for later tracking	Scenario 2: Adding Match to Interests
US-UNFOLLOW-MATCH	User Story	Remove a match from the followed list	Scenario 3: Removing Match from Interests
US-MATCH-DETAILS	User Story	View detailed information about a specific match	Scenario 4: Viewing Match Details
US-VIEW-FOLLOWED	User Story	View only the matches currently being followed	Scenario 5: Viewing Followed Matches
FR-DATA-STORE	Functional Req	Store match information (teams, date/time, venue, league, status)	FR1: Match Data Management
FR-DATA-API	Functional Req	Retrieve match data from a reliable sports data source via API	FR1: Match Data Management
FR-USER-FOLLOW	Functional Req	Authenticated users can follow/unfollow any match	FR2: User Interest Tracking
FR-FILTER-LEAGUE	Functional Req	Filter matches by league/competition	FR3: Match Filtering and Search
NFR-FRESH-LIVE	Non-Functional Req	Live match scores updated at least every minute during active matches	FR5: Data Freshness
SC-PERF-LOAD	Success Criterion	Match list loads in under 2 seconds for 95% of requests (cached)	SC2: Performance

ENT-MATCH	Entity	Match: matchId, homeTeam, awayTeam, dateTime, venue, league, status, scores	Match
ENT-TEAM	Entity	Team: teamId, name, logo	Team
ENT-USER	Entity	User: userId, username, email	User (if authentication implemented)
ENT-USERFOLLOW	Entity	UserFollow: Links a user to a match they follow (userId, matchId)	UserFollow (join entity)
AS-API-Avail	Assumption	Assume access to a reliable football match data API	A1: Data Source
C-NO-NOTIF	Constraint	Push notifications are considered a post-MVP enhancement	A3: Scope Limitations

3.2 Security Rules

- ****Authentication****: Access to "Follow" and "My Matches" features requires a valid session via third-party OAuth (Google, Facebook, Apple).
- ****Session Persistence****: User follow relationships must be persisted across sessions via the `ENT-USERFOLLOW` entity.

3.3 Data Models

- ****Match (ENT-MATCH)****: Primary entity containing match metadata and real-time scores.
- ****Team (ENT-TEAM)****: Reference entity for team names and branding.
- ****League (ENT-LEAGUE)****: Categorization entity for grouping matches.
- ****User (ENT-USER)****: Account entity for authenticated fans.
- ****UserFollow (ENT-USERFOLLOW)****: Join table managing the many-to-many relationship between Users and Matches.

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Gap	Priority	Remediation Advice
Goals & Objectives	HIGH	Define the high-level business goals and success KPIs for the project to align the team on 'Why' this is being built.

4.2 Remediation & Workflow

1. ****Strategic Alignment****: Conduct a stakeholder workshop to define the "Goals & Objectives" section.
2. ****Technical Selection****: Evaluate and select a specific Football Data API provider based on cost and reliability.
3. ****Auth Protocol****: Finalize the specific OAuth flow for the chosen third-party providers.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
API	TECHNICAL_STACK	FR-DATA-API	the external sports data interface used to retrieve schedules, live scores, and league information
MVP	TECHNICAL_STACK	A3: Scope Limitations	the initial version of the product focused on tracking interests and viewing data, excluding social features and push notifications
Scenario	BUSINESS_DOMAIN	Scenario 1: Viewing Match List	a described user journey mapping specific interactions and expected outcomes for a football fan
UserFollow	BUSINESS_DOMAIN	ENT-USERFOLLOW	the join entity associating a specific authenticated account with a targeted game via a timestamped relationship